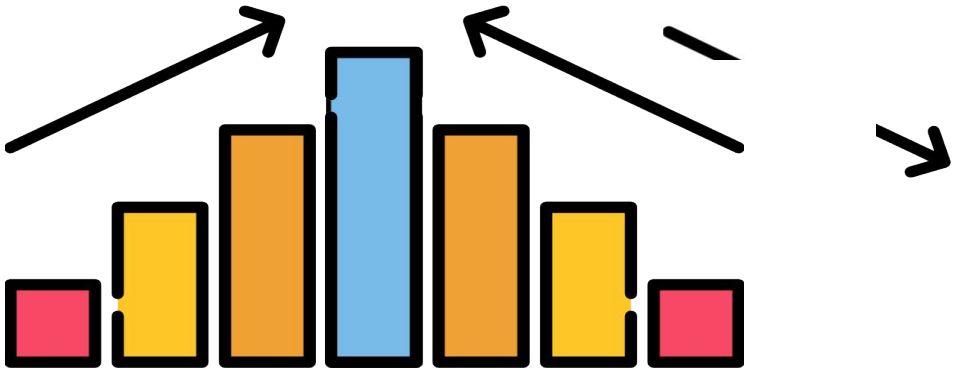
Monotonicity



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Monotonic stack

- A stack whose elements are **monotonically** increasing or decreasing.
- Useful when we're looking for the next larger/smaller element.
- For a mono-decreasing stack:
    - we need to pop smaller elements before pushing.
    - By continually removing smaller elements and retaining greater ones, the mono-decreasing stack refines the result to achieve the maximum lexicographical greatness possible.

luu

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Practice

Problem

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Solution

```python
class Solution:
    def nextGreaterElement(self, nums1: List[int],
    nums2: List[int]) -> List[int]:
        stack = []
        res = defaultdict(lambda: -1)
        for num in nums2:
            while stack and stack[-1] < num:
                res[stack[-1]] = num
                stack.pop()

            stack.append(num)

        return [res[num] for num in nums1]
```

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Practice

Problem

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Solution

```python
class Solution:
    def dailyTemperatures(self, temperatures: List[int]) -> List[int]:
        stack = []
        ans = [0] * len(temperatures)

        for i in range(len(temperatures)):
            while stack and temperatures[stack[-1]] < temperatures[i]:
                ans[stack[-1]] = i - stack[-1]
                stack.pop()

            stack.append(i)

        return ans
```

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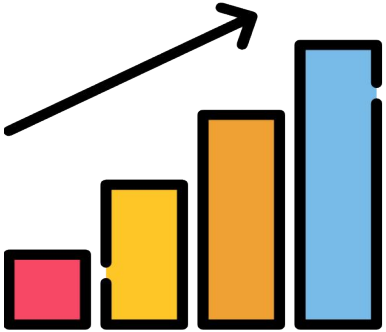
Monotonic Stack Application



- It gives you how far a value spans as a **maximum** or **minimum** in the given array.

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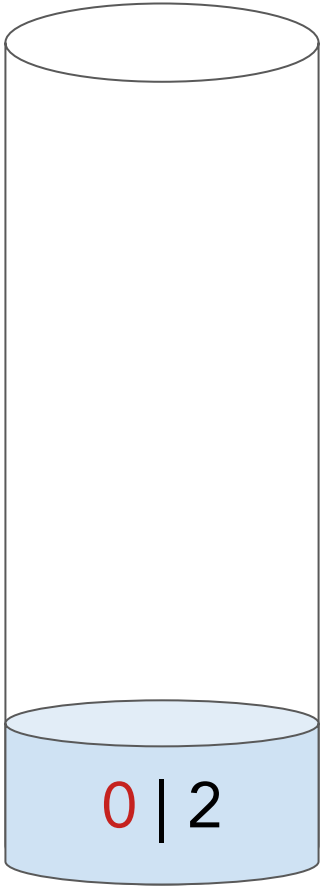
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing stack



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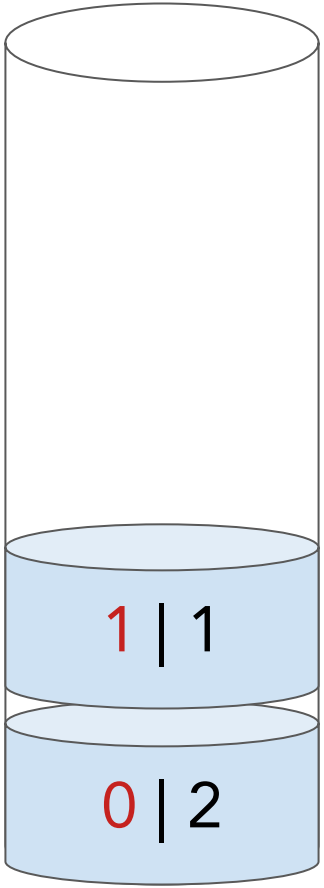

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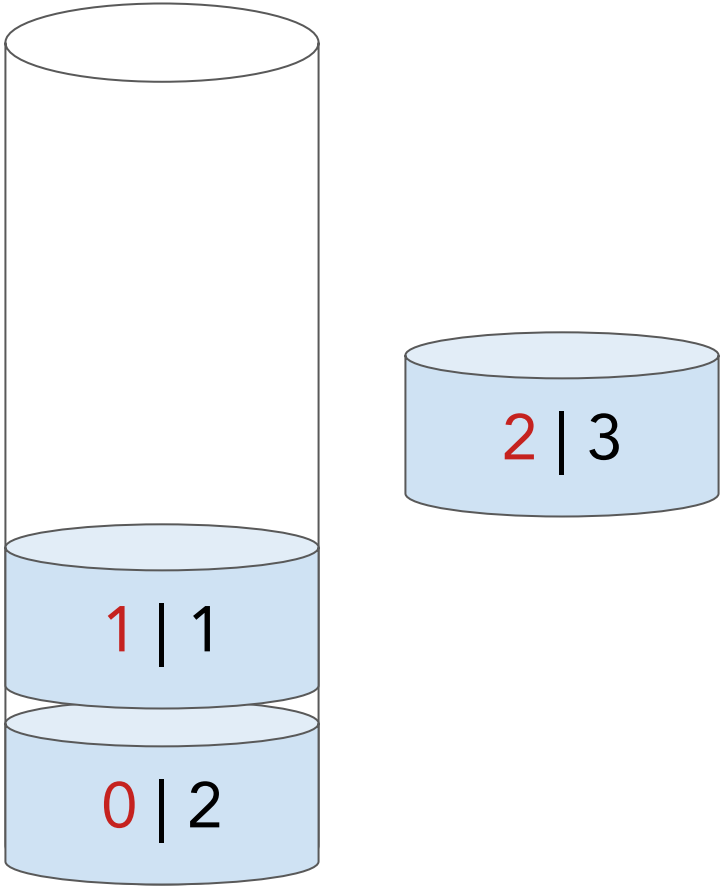

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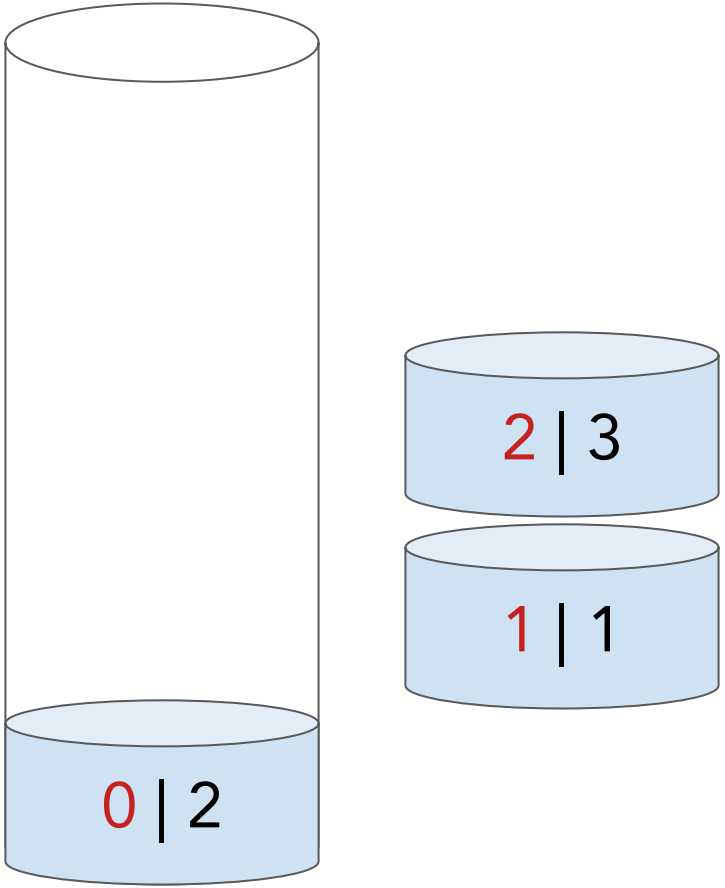

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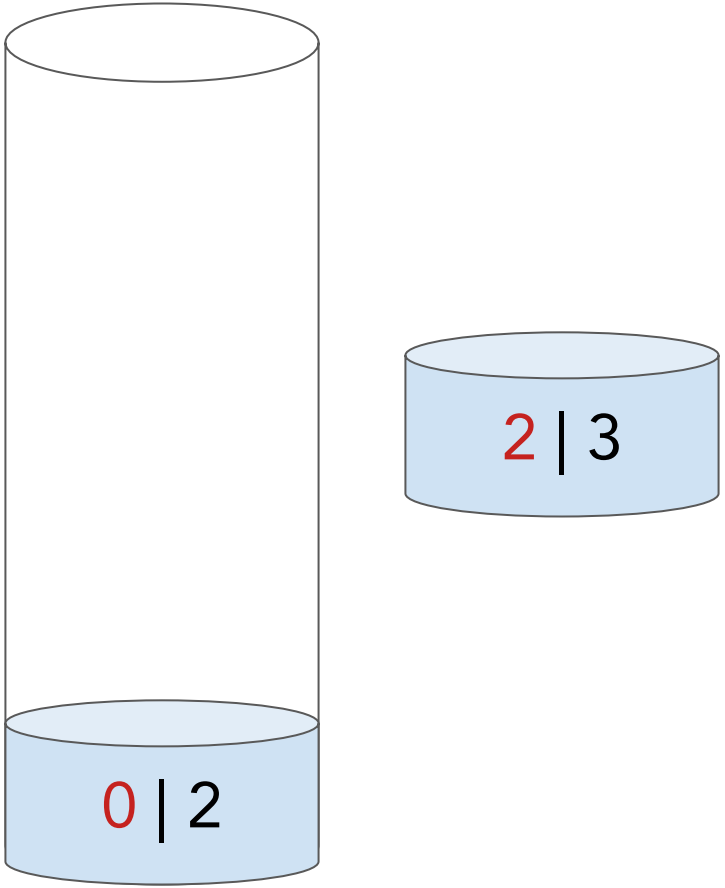

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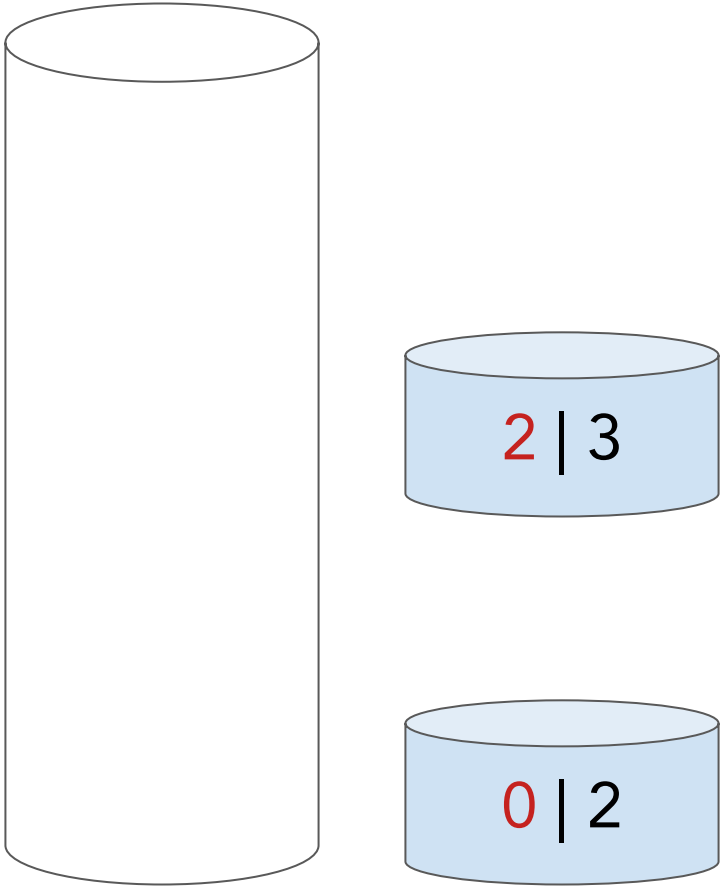

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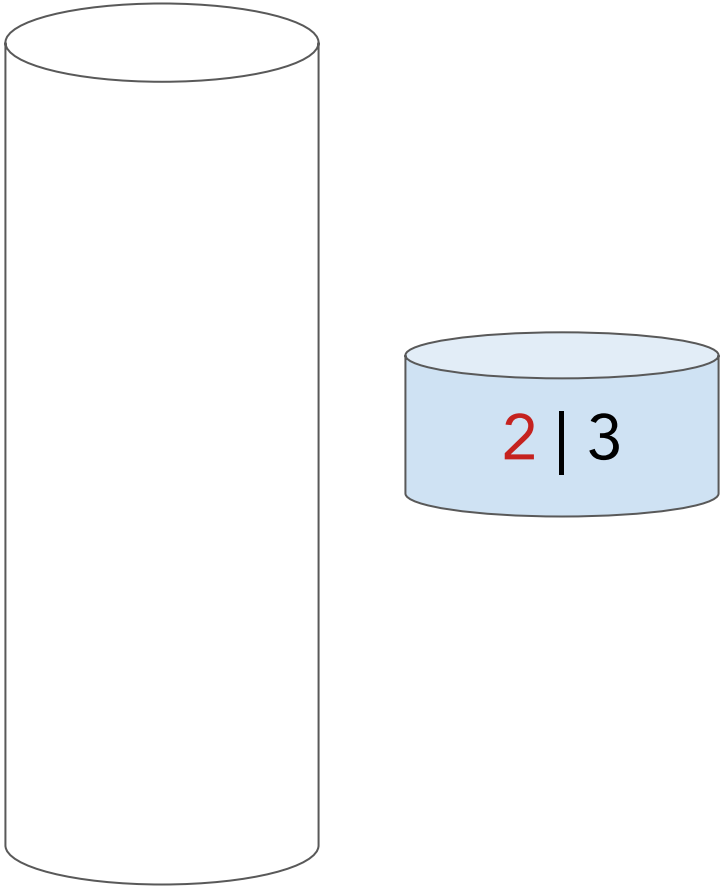

Monotonically decreasing stack

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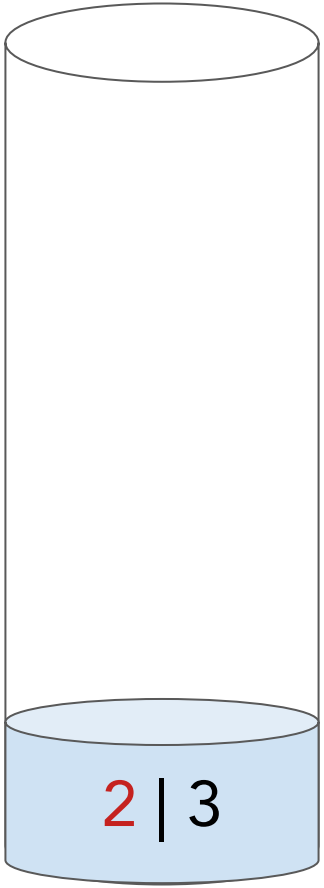

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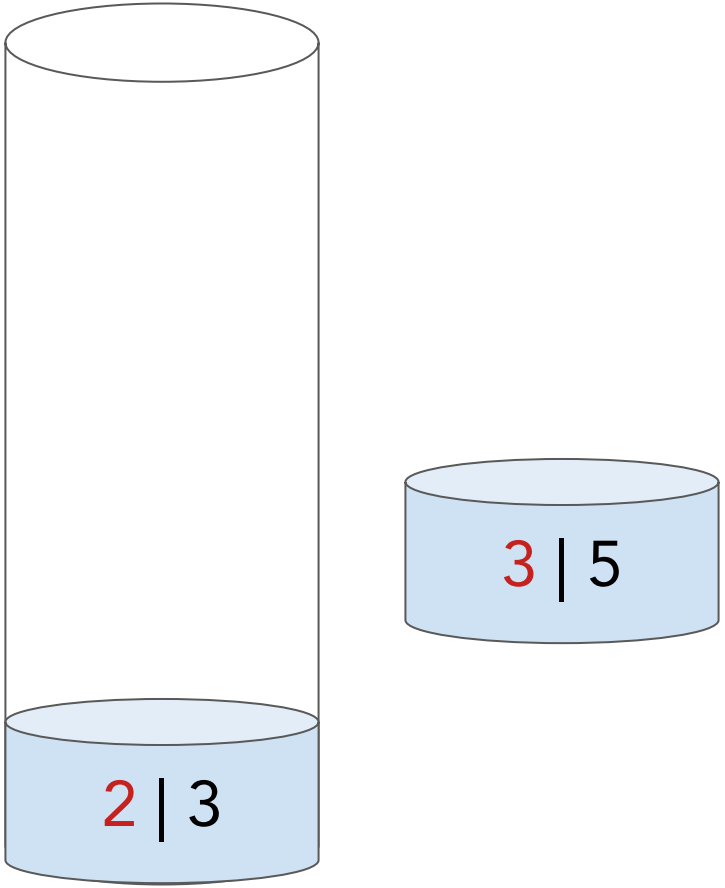

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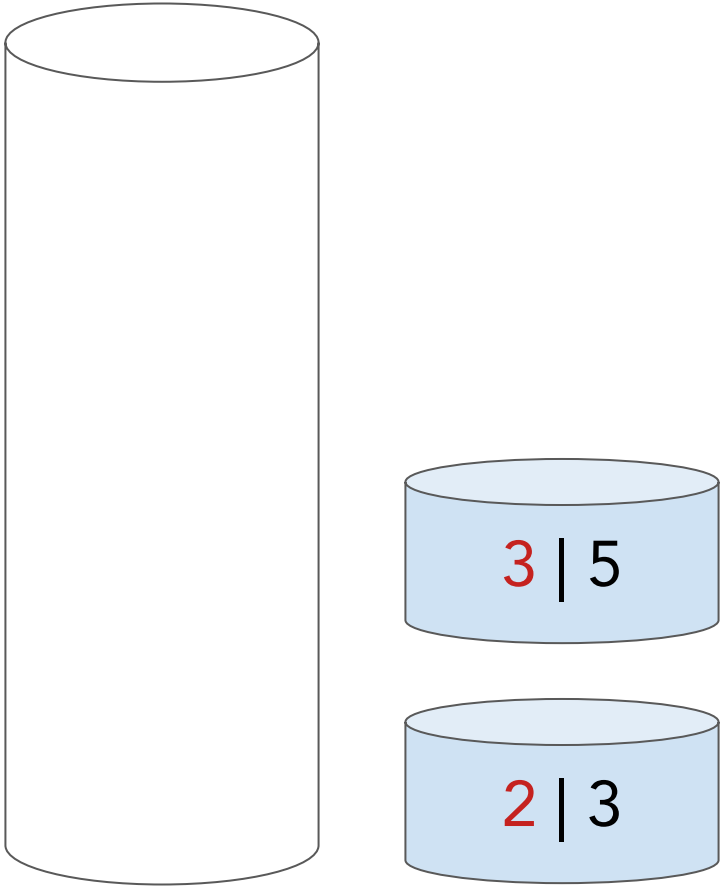

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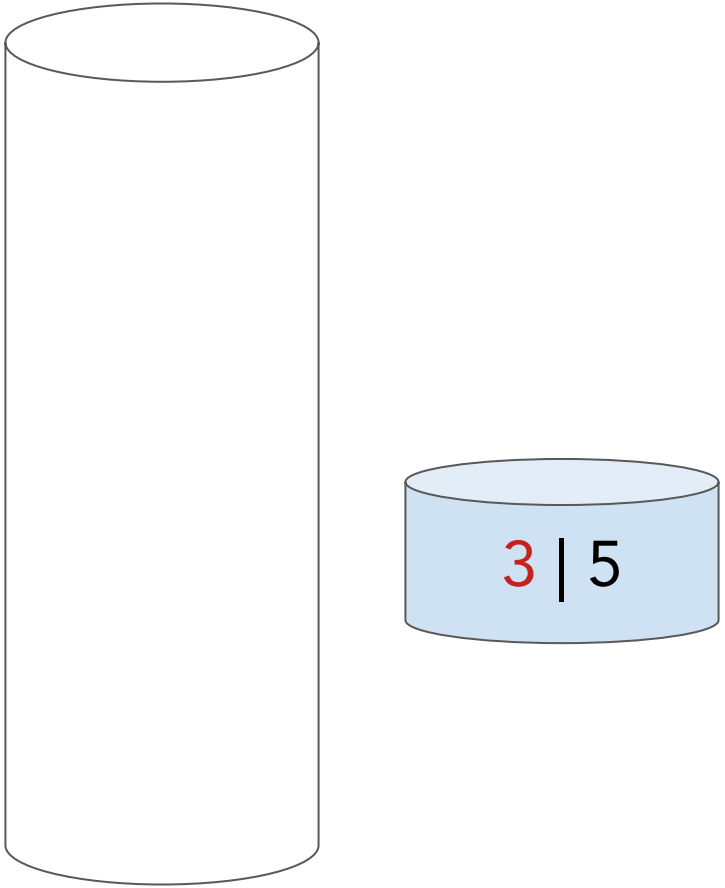

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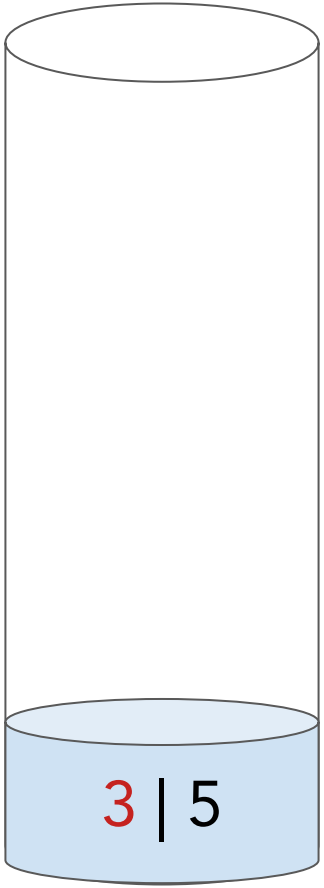

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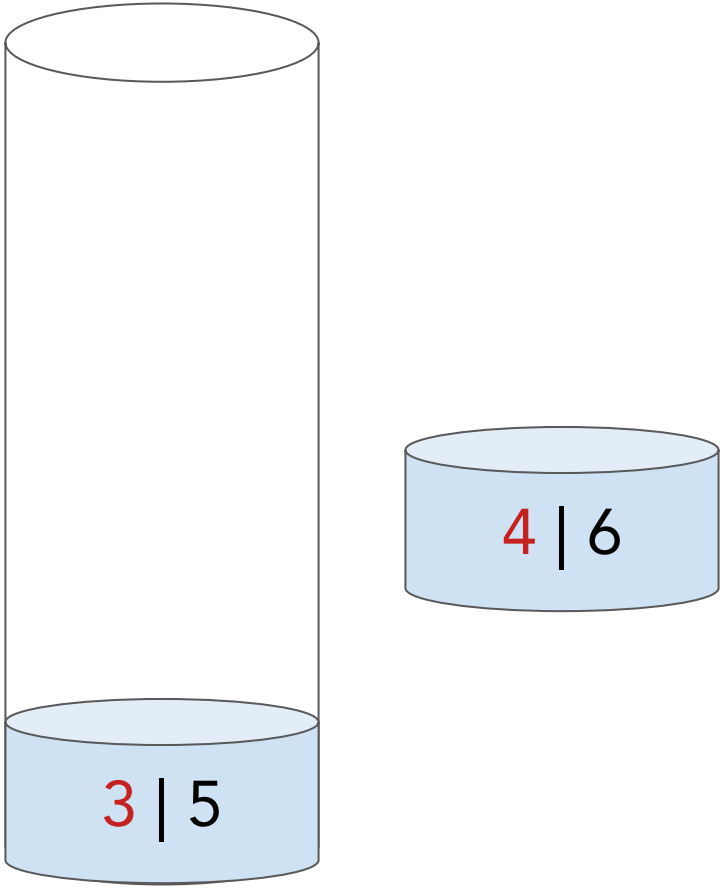

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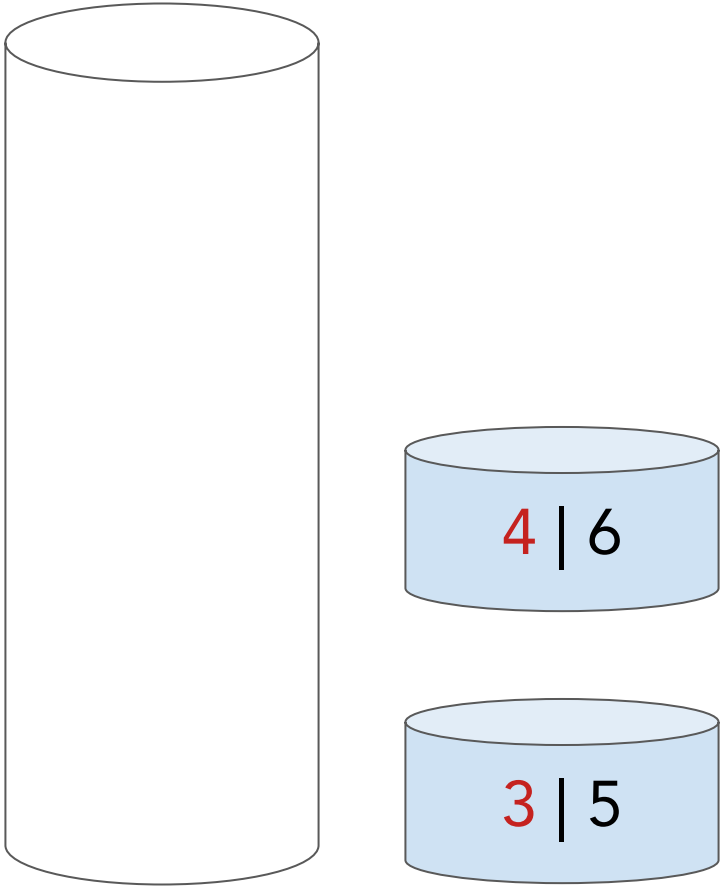

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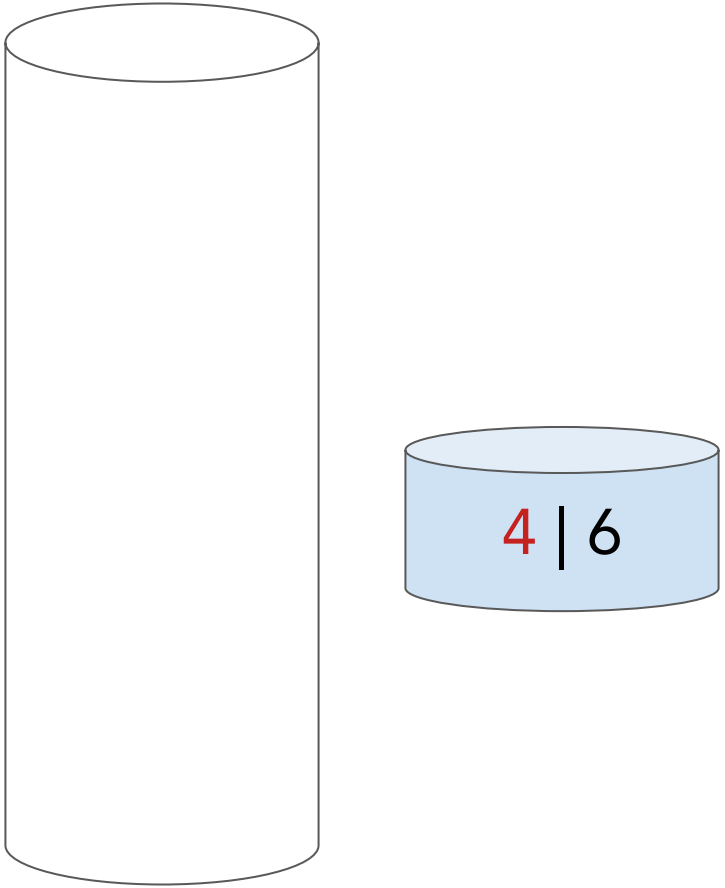

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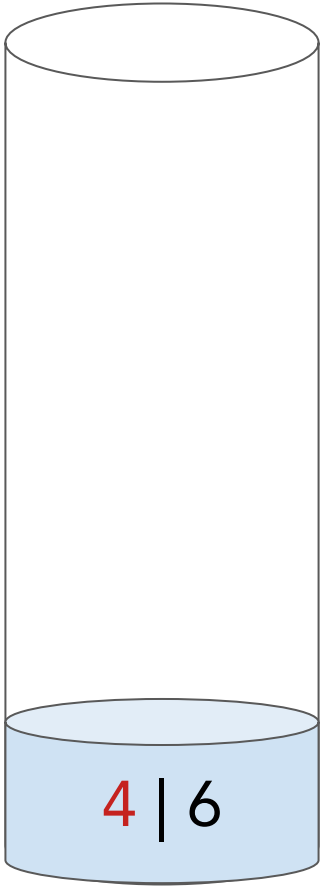

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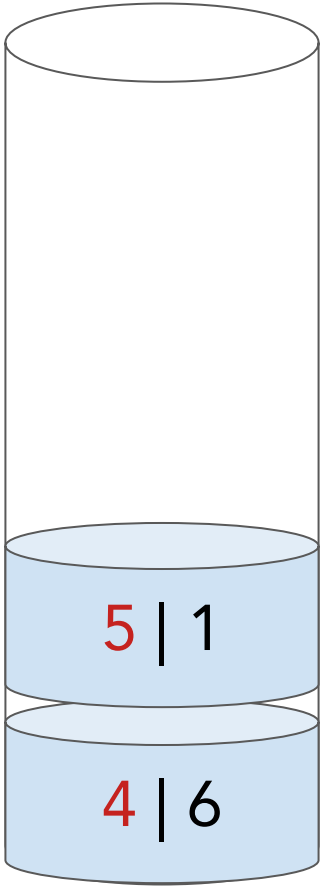

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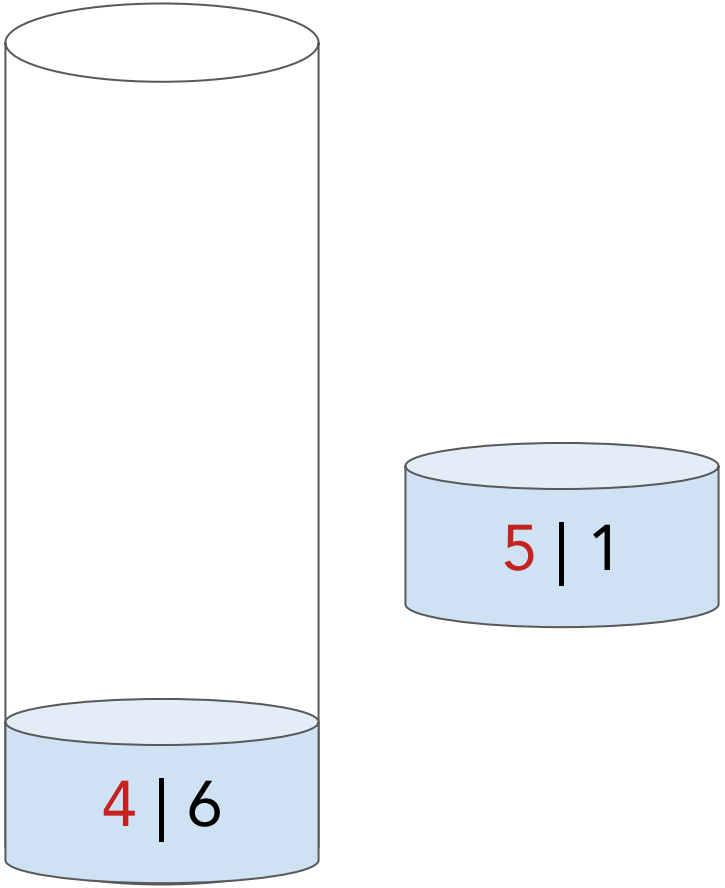

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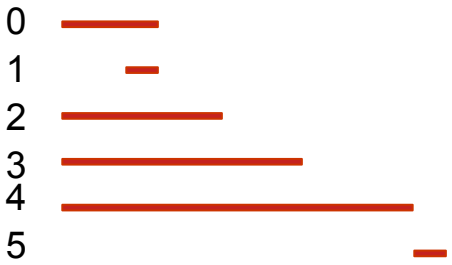

Index Spans:

1: 1 <= … < 2

0: 0 <= … < 2

2: 0 <= … < 3

3: 0 <= … < 4

5: 4 <= … < 5

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Monotonic Queue

- A queue whose elements are monotonically increasing or decreasing.
- For a mono-decreasing Queue:
    - To push an element e, the process begins with the rear element, where elements smaller than e are sequentially popped out.

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Pair Programming

Problem

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Solution

- Create a min queue to track the minimum element in our window.
- Create a max queue to track the maximum element in our window.
- If max - min is greater than limit, shrink the window
    - If number about to be popped is in either of the queues, pop it.

```python
from collections import deque

class Solution:
    def longestSubarray(self, nums, limit):
        min_queue = deque()
        max_queue = deque()
        start = 0
        max_size = 0

        for end in range(len(nums)):
            # add to min queue
            while min_queue and min_queue[-1] > nums[end]:
                min_queue.pop()
            min_queue.append(nums[end])

            # add to max queue
            while max_queue and max_queue[-1] < nums[end]:
                max_queue.pop()
            max_queue.append(nums[end])

            while max_queue[0] - min_queue[0] > limit:
                if nums[start] == min_queue[0]:
                    min_queue.popleft()
                if nums[start] == max_queue[0]:
                    max_queue.popleft()
                start += 1

            max_size = max(max_size, end - start + 1)

        return max_size
```

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Time and Space Complexity

- The time and space complexity for monotonic stack and queue operations are the same as stack and queue operations.
  - **Why**?

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Check point

Pitfalls & Opportunities

- Be careful of how to handle equality
    - Should we pop elements in the monotonic stack/queue that are equal?
- Check if stack/queue is empty before accessing/removing
- For greater problems, usually use a monotonically increasing stack
- For smaller problems, usually use a monotonically decreasing stack

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Practice Questions

Stacks

- Valid Parentheses
- Simplify Path
- Evaluate Reverse Polish Notation
- Score of parenthesis
- Backspace String Compare

Queues

- Number of recent calls
- Find consecutive integers
- Design Circular Deque
- Implement Queue using Stack
- Shortest subarray with sum at least K

Monotonic

- Car Fleet
- Remove duplicates
- Sum of subarray minimum
- Remove k digits
- 132 Pattern

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Resources

A comprehensive guide and template for monotonic stack based problems

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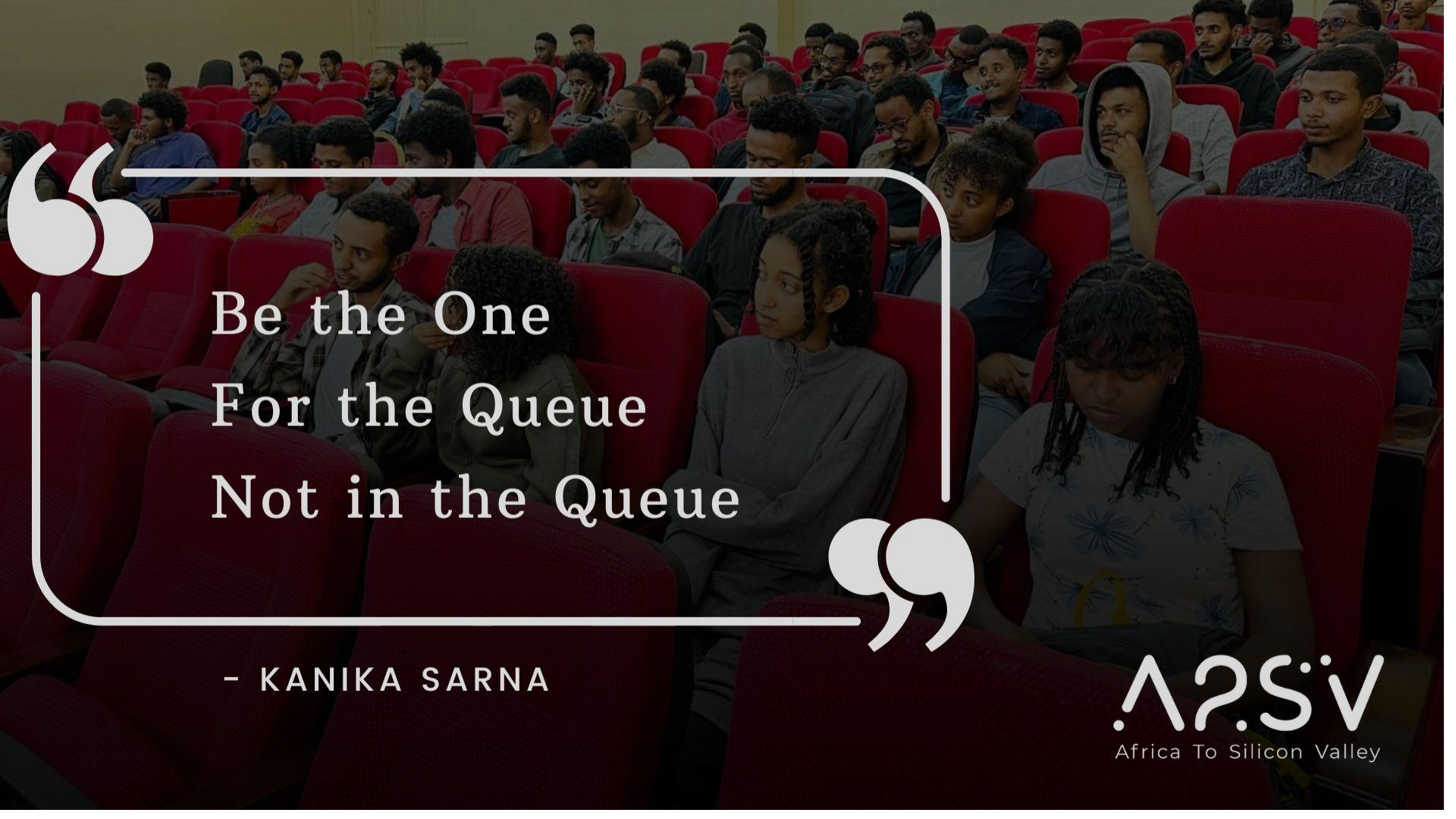
Be the One
For the Queue
Not in the Queue
- KANIKA SARNA
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